



Laboratory Work 5

Optimal Observer (kalman filter) and LQG controller

Lab Goal: To design an optimal observer using kalman filter and the LQG controller

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Lab Variant: 28

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1. Introduction

In this laboratory work we will design an optimal observer to estimate the system states using the kalman filter and the LQG controller for the controller. The plant is in the form,

$$\begin{aligned}\dot{x} &= Ax + Bu + Gw \\ y &= cx + v\end{aligned}$$

Where w and v are white noises with zero expected values and autoregressive functions,

$$M\{w(t)w^T(\tau)\} = w\delta(t - \tau), \quad M\{v(t)v(\tau)\} = v\delta(t - \tau)$$

Our objective is to design a state observer for the system where,

$$|x(t) - \hat{x}(t)| \leq \Delta \quad \forall t \geq T$$

Where Δ and T are the maximum steady state error and the transient time

The cost function to calculate the optimality of the system is as,

$$J = M\{e_H^T e_H\}$$

Where e_H is

$$x(t) - \hat{x}(t)$$

The given data for building the system is as,

$$A = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad C = [1 \ 0]$$

$$W = \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix}, \quad V = 3$$

$$Q = \begin{bmatrix} 9 & 0 \\ 0 & 7 \end{bmatrix}, \quad R = 1$$

Using the data we will build the observer,

$$\dot{\hat{x}} = A\hat{x} + bu + L(y - C\hat{x})$$

For that we need to calculate the matrix L first using the Riccati equation

$$AP + PA^T + GWG^T - PC^TV^{-1}CP = 0$$

$$L = PC^TV^{-1}$$

We will use the Matlab's continuous time Algebraic Riccati equation to calculate the Matrix L using our existing data

2. Modeling



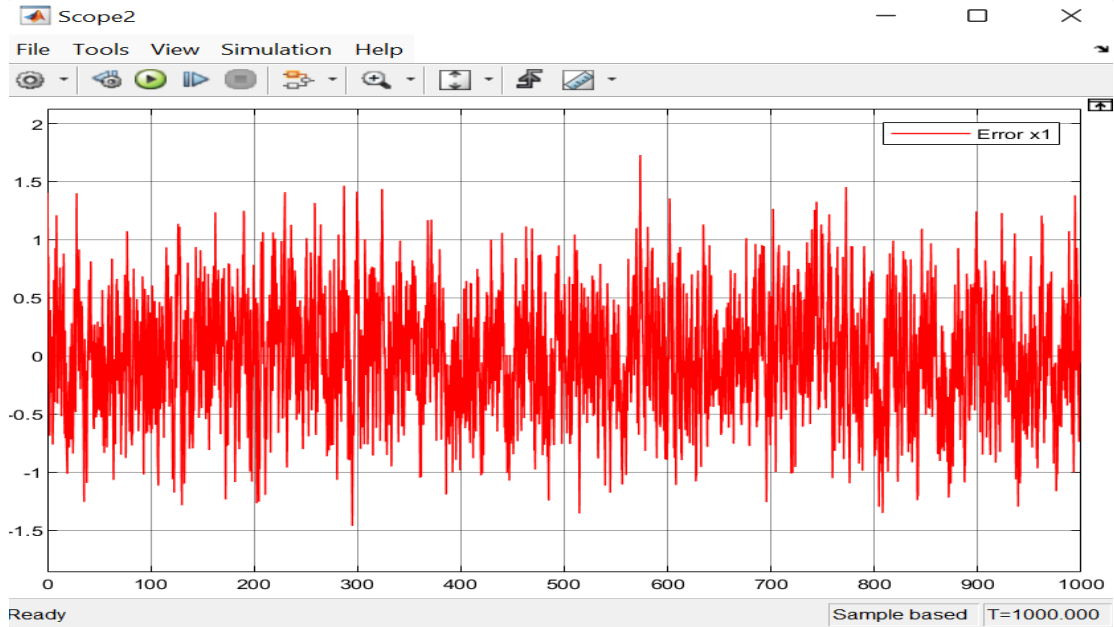


Figure 3. Error eh1

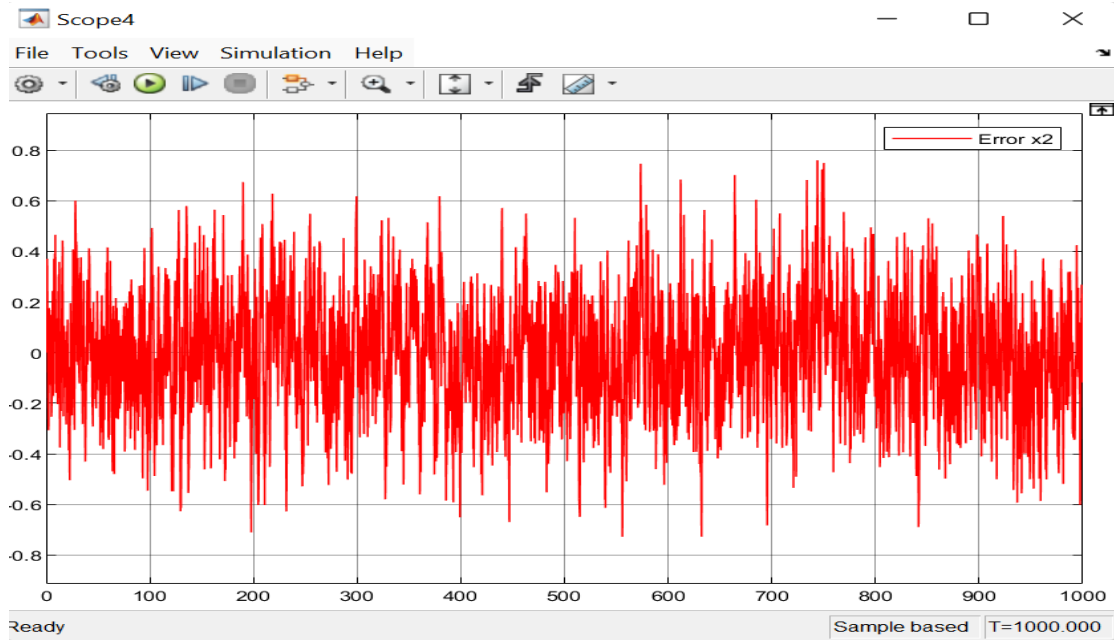


Figure 4. Error eh2

We can see some minor error in the observation of states but that is due to the white noise present in the system, the observer is working correctly as the observer error is converging to 0 without the noises

Now let's try changing the values of L,W and V one by one to analyze the results and their effects on the system.

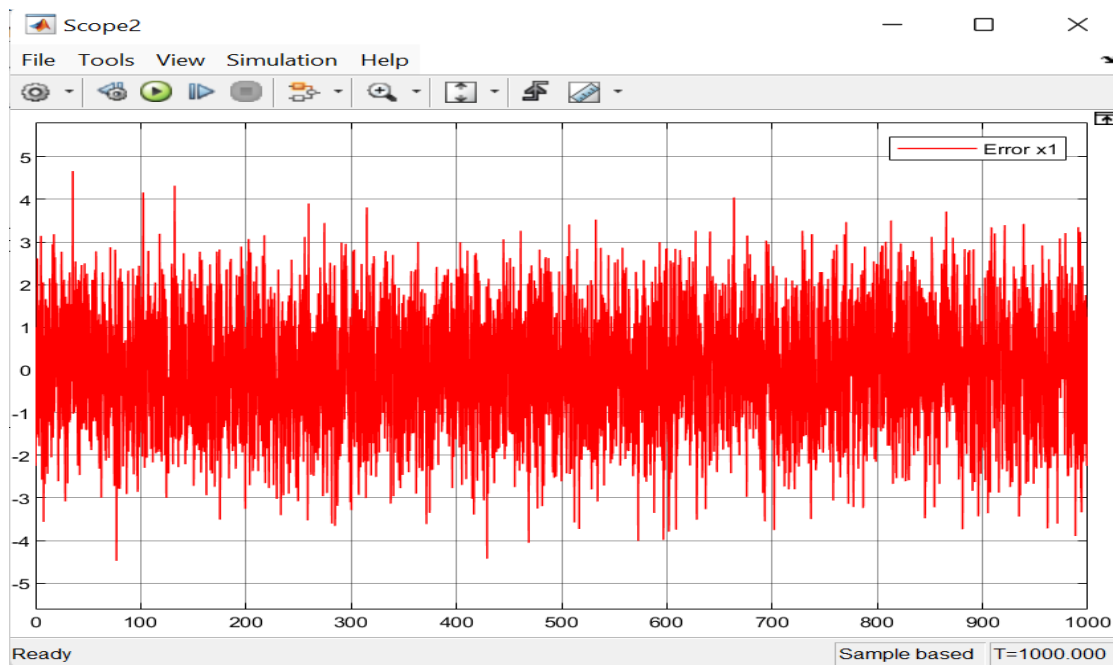


Figure 5. Error eh1 after changing L

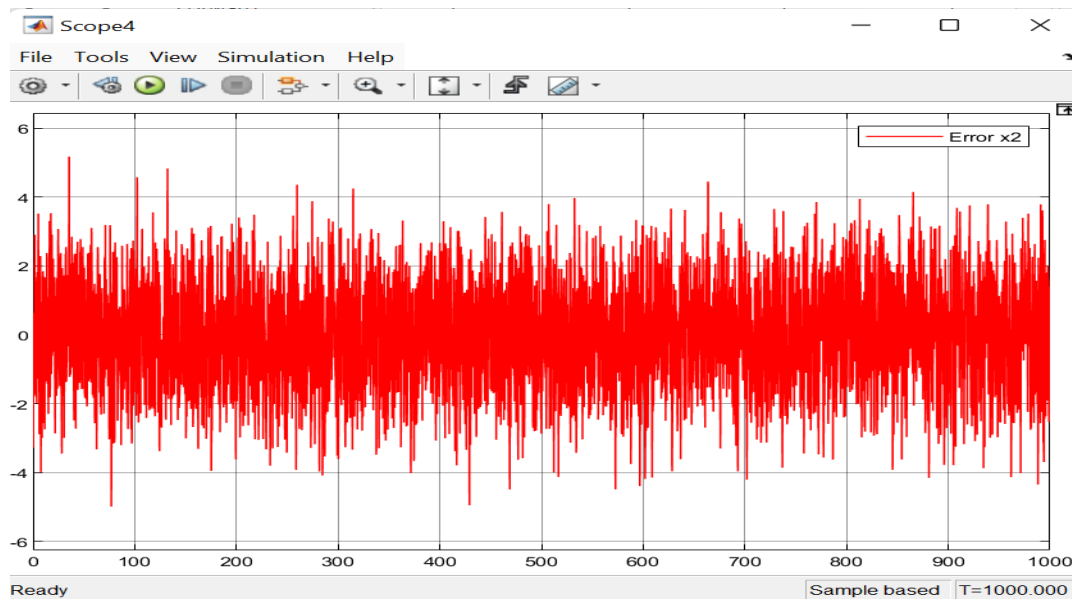


Figure 6. Error eh2 after changing L

We can observe that by changing the Matrix L negligibly, the error of the system has increased a bit but the stability is still preserved

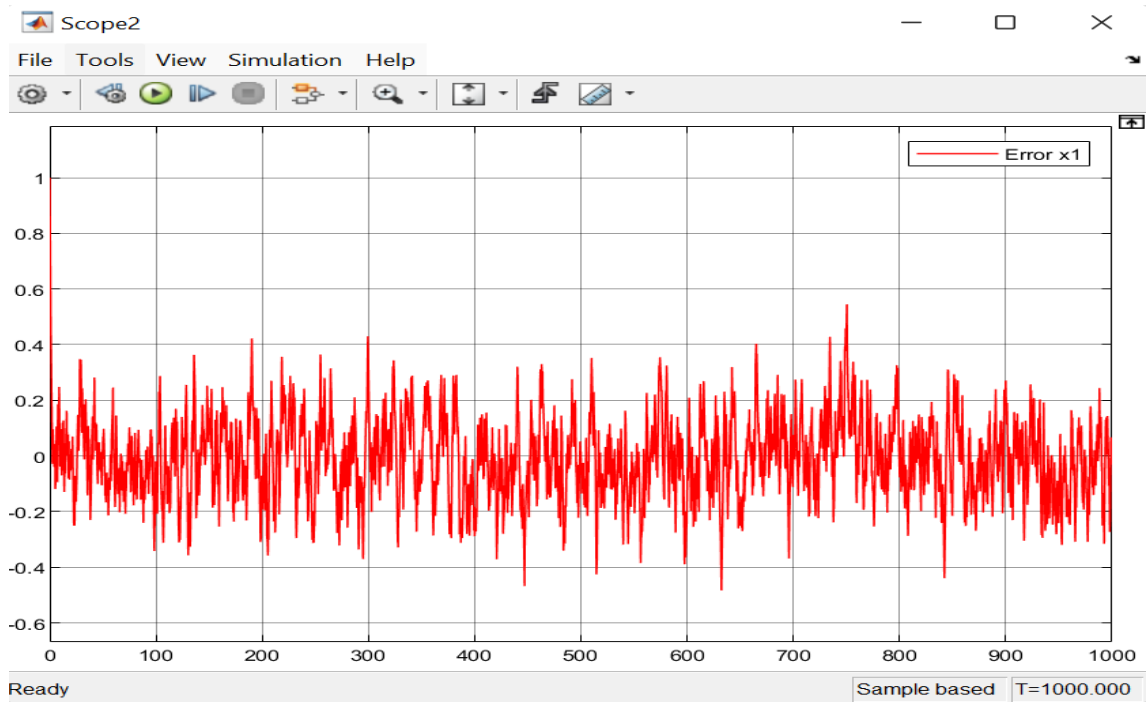


Figure 7. Error eh1 by changing W

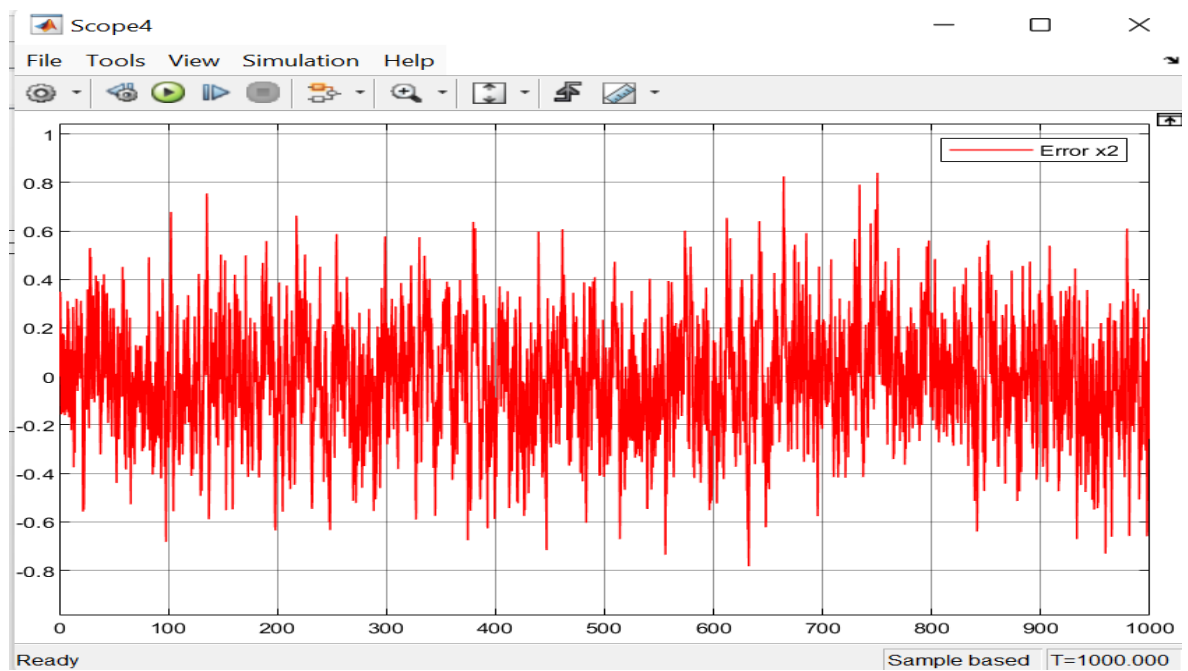


Figure 8. Error eh2 by changing W

By changing W to a different positive defined Matrix we can observe that the error of the system is reduced as compare to before

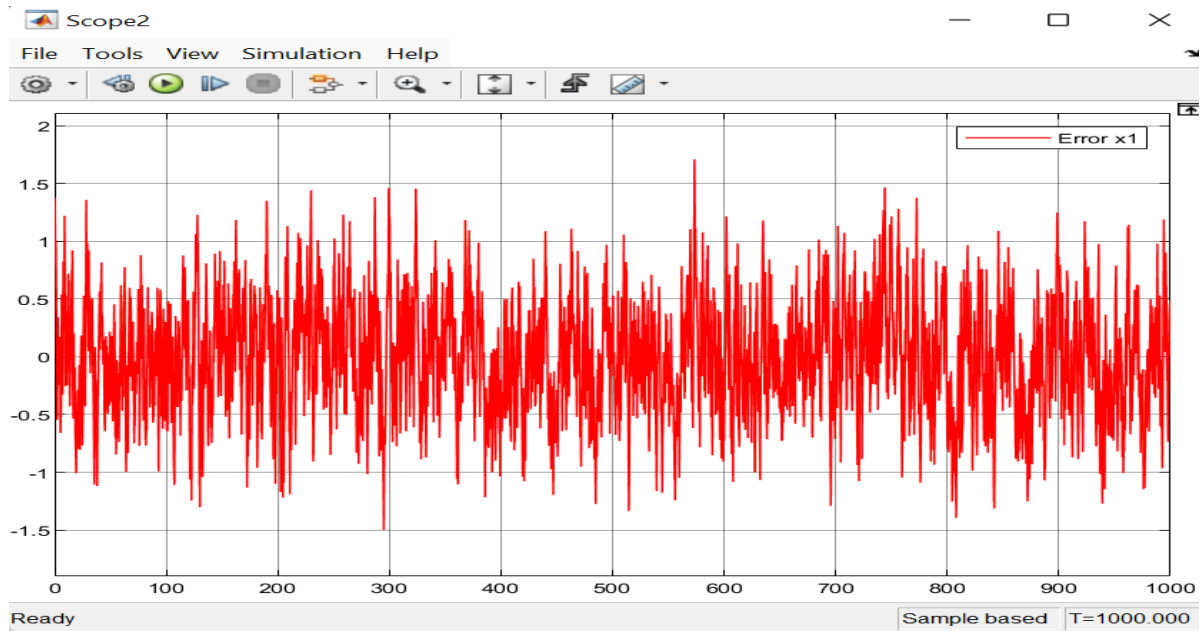


Figure 9. Error eh1 by changing V

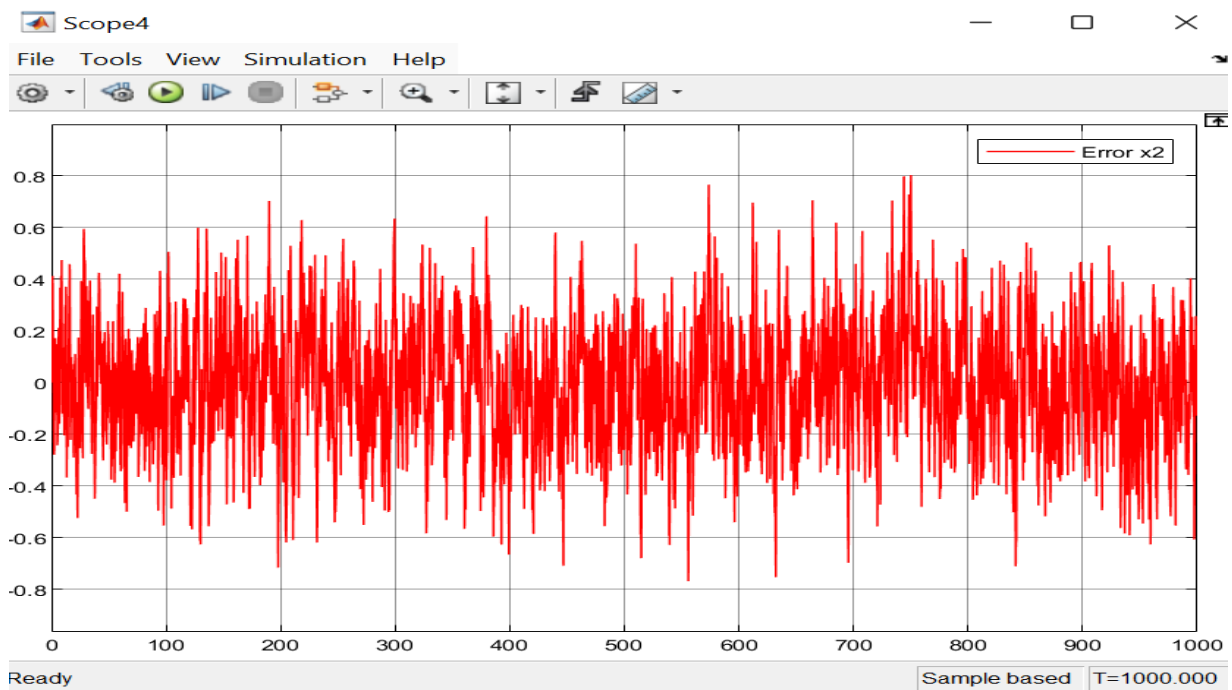


Figure 10. Error eh2 by changing V

By changing the value of V there isn't much difference in the system since the value of white noise is not very big, so it has a negligible effect

Now we will calculate the optimal controller for our original system using the LQR and the data from the previous lab work to find the gain K and then check the output for the optimal system, let's calculate K and for the controller as $u = -K\hat{x}$

Where $\hat{x}$ will be taken from the kalman filter observer. Using the lqr solver from matlab, the K matrix is found to be,

$$[2.127; 1.7883]$$

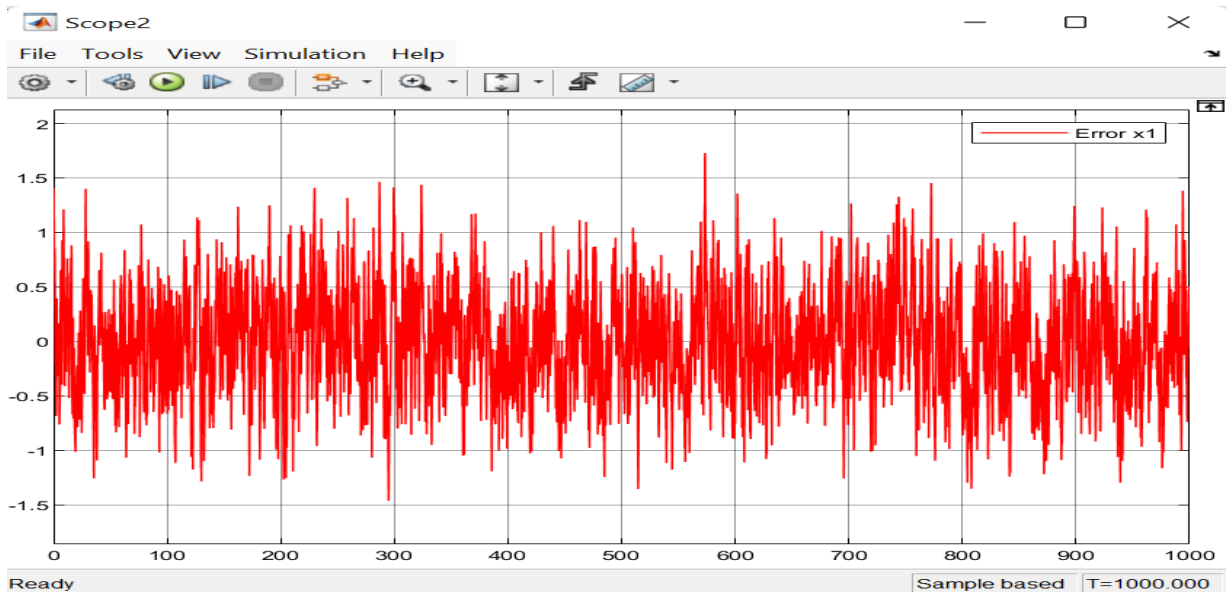


Figure 11. Error x1 with optimal controller

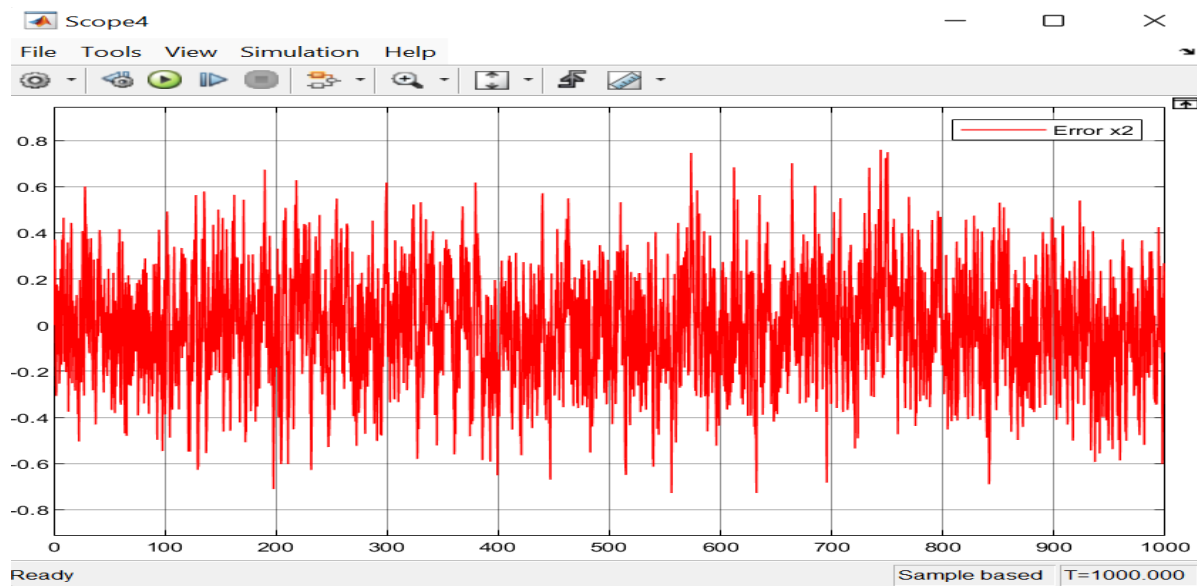


Figure 12. Error x2 for optimal controller

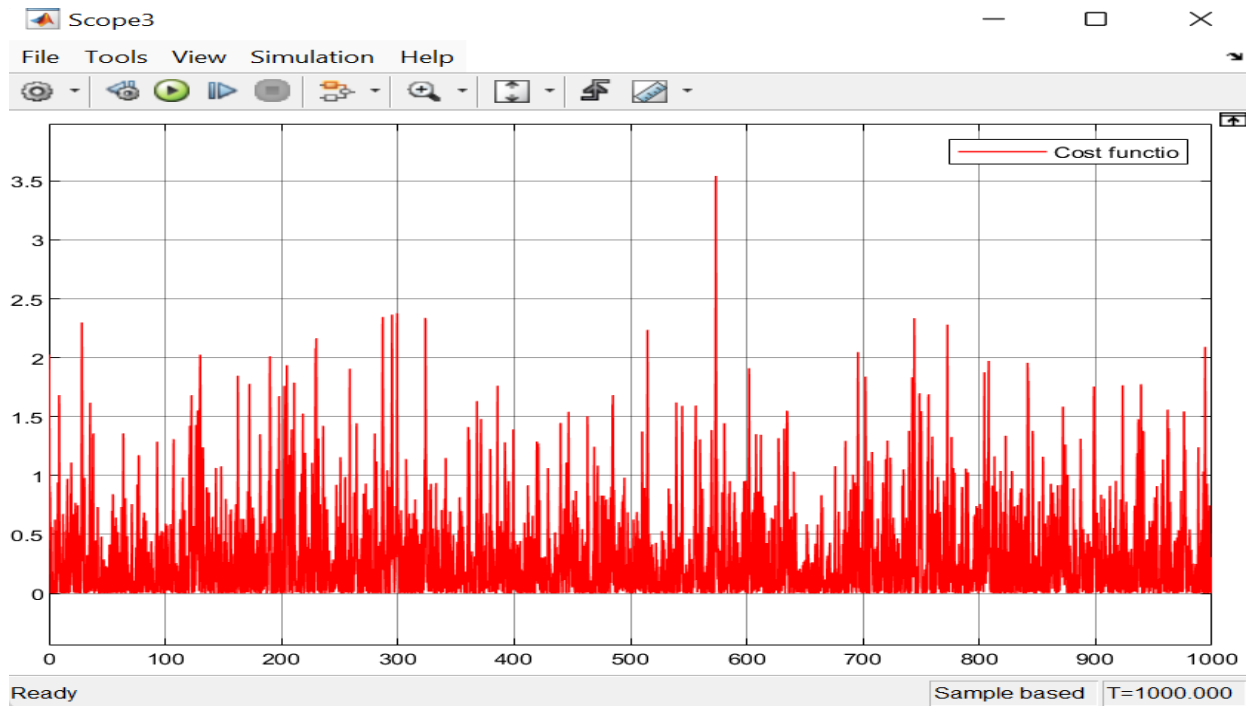


Figure 13. Cost function J for optimal controller

4. Conclusion

In this laboratory we have used the kalam filter to calculate the optimal observer for the states of our system and analyzed the effects of different parameters, We have used white noises with gaussian distribution with an expected value of zero, furthermore we have derived an optimal controller for our system that regulates the system to observe the states with the minimum cost as possible.